

MAC21 — Prediction Sheet

Numerical Techniques and Differential Equations · B.E. Semester II · Ramaiah Institute of Technology · SEE, 100 marks, 3 hours

Built from 6 papers: the Department of Mathematics Model Question Paper I (MP) · MAC21 Sep/Oct 2023 (M23) · MAC21 Jul/Aug 2024 (M24) · MAM21 Jul/Aug 2024 (MAM24, Units I-IV) · MAE21 Sep/Oct 2023 (MAE23, Units I-III) · 25MAC21 Jun/Jul 2026 (P26, partial — different scheme).
Everything off your syllabus has been stripped out — no multiple integrals, no Laplace transforms, no Beta/Gamma functions, no PDEs, no Regula-Falsi, no LU decomposition, no Weddle's rule. Only MAC21 content appears below.

How to read the repeat counts. Coverage differs by unit, so a count is always *out of the papers that actually carry that unit*. Units I-III → **6 papers** · Unit IV → **5 papers** (MAE23's Unit IV is Laplace) · Unit V → **3 papers** (only MP, M23, M24 — the 2026 scheme deleted this unit). A "3×" in Unit V is therefore a clean sweep, not a weak signal.

1 The paper pattern

ELEMENT	RULE — IDENTICAL IN EVERY PAPER CHECKED
Total	100 marks, 3 hours
Sections	5 units. Unit I → Q1 / Q2 · Unit II → Q3 / Q4 · Unit III → Q5 / Q6 · Unit IV → Q7 / Q8 · Unit V → Q9 / Q10
Choice	Answer ONE full question from each unit . Internal choice, always exactly 2 options. You attempt 5 questions total.
Marking	Every question is a) 02 + b) 04 + c) 07 + d) 07 = 20 marks . Never varies.
Part (a) — 2 mk	Always pure recall . A definition, a formula to state, or "write the steps involved in...". <i>Never</i> a calculation, in any paper.
Part (b) — 4 mk	Short calculation: one iteration, a small proof, a P.I., a definition-with-example.
Parts (c),(d) — 7 mk	Full problems. Typically one standard technique + one applied word problem.
CO tag	Unit <i>n</i> is always tagged CO <i>n</i> . Mapping is 1:1 and stable.

What this hands you: the five 2-mark parts are **10 marks of pure memorisation**, and you get to pick the easier of two in each unit. The bank of definitions ever used is small — it is listed in full in section 6.

2 Which question to answer in each unit

The two options in a unit are not random. Each carries a recurring *combination* of topics, stable across years. Here is what each side reliably holds.

UNIT	ANSWER	WHY — WHAT THAT SIDE RELIABLY BUNDLES	CONFIDENCE
I	Q2 (usually)	Q2 holds the "expand ... about (1, $\pi/2$)" 7-marker (4×) and the maxima-minima 4-marker (3×). Q1 holds the Newton-Raphson system (5× at slot 1c) + Lagrange multipliers. Prepare both — each side has one near-certain 7-marker. Decide by the (d) parts.	Medium — genuinely split
II	Q4	Modified Euler sits at exactly 4c in 4 of 5 papers . Q4 also carries the circuit-DE 2-marker and a bungee/self-orthogonal item. Q3 leans Taylor's method + cooling + RK4.	High
III	Q5	In all three exact-match papers Q5 = $1/(D+a)$ proof [4] + Cauchy [7] + variation of parameters [7] = 18 of 20 marks from three techniques you can fully prepare. Q6 is the Legendre/miscellaneous side — far less stable.	Very high

IV	Q8	Q8 bundles Lagrange inverse formula [2] + missing terms [4] + divided difference [7] + grouped frequency [7] — the four most repetitive items in the unit. Missing terms sits at 8b in 3 of 3 papers.	Very high
V	Q10	Q10 = definition [2] + Gauss-Seidel [4] + Gauss elimination / consistency [7] + system of ODEs by matrix method [7] . That last one is at slot 10d in 3 of 3 papers — always the final question of the exam. But learn all of Unit V; it is only ~6 techniques.	Very high

3 TIER 1 — Lock. Same question, same values.

TIER 1 — expect these verbatim

Each has appeared **2–4 times with the same numbers**. Learn the worked answer, not the method. If it shows up you write it from memory. **Total on the table: 60 marks**. Realistically 30–40 of these land.

QUESTION	MK	UNIT	SLOT	REPEATS	SEEN IN	APPEARS ALONGSIDE
Prove $[1/(D+a)]X = e^{-ax} \int X e^{ax} dx$ where $D = d/dx$	4	III	Q5(b)	4×/6	MP, M23, M24, MAE23	Steps for Cauchy's LDE [5a] · Cauchy's equation [5c] · variation of parameters [5d]
Expand $xy + \cos(xy)$ about $(1, \pi/2)$ up to second degree terms (variants: $xy + \sin(xy)$, $xy^2 + \cos(xy)$ — the point $(1, \pi/2)$ never changes)	7	I	Q2(c)	4×/6	M23, MAM24, MAE23, MP	Maclaurin/Taylor statement [2a] · $xy(1-x-y)$ maximum [2b] · single-equation N-R [2d]
Newton-Raphson on the system $x^2 + y^2 = x$, $x^2 - y^2 = y$, $x_0 = 0.8$, $y_0 = 0.4$, two iterations (P26 rewrites it as $x = x^2 + y^2$, $y = x^2 - y^2$ — same system)	7	I	Q1(c)	4×/6	M23, M24, MAM24, P26	Geom. interpretation of N-R [1a] · series expansion [1b] · Lagrange multipliers [1d]
Define linear and non-linear differential equations with example	2	III	Q6(a)	3×/6	M23, MAE23, P26	Legendre's equation [6c] · IVP/BVP [6b or 6d]
Show $f(x, y) = xy(1-x-y)$ is maximum at $(1/3, 1/3)$	4	I	Q2(b)	3×/6	MP, MAM24, MAE23	Expand about $(1, \pi/2)$ [2c] · Maclaurin statement [2a]
Find the missing terms: $x = 45, 50, 55, 60, 65$; $y = 3.0, -, 2.0, -, -2.4$	4	IV	Q8(b)	3×/5 identical table 2×	MAM24, P26 (same data); M24 (other data)	Lagrange inverse formula [8a] · grouped frequency [8c] · Simpson's 3/8 [8d]
Coffee at 92°C, room 24°C, cooled to 80°C in 1 minute — time to reach 65°C? (M24 runs 90 / 24 / 75 / 60)	7	II	Q3(d)/Q4(d)	3×/6 identical 2×	MP, P26 (same); M24 (variant)	Taylor's series method [3c] · Euler's method [3b] · RK4 formula [3a]
Cubic through $(2, 4)$ $(4, 56)$ $(9, 711)$ $(10, 980)$ by Newton's divided difference; estimate y at $x = 1.5$	7	IV	Q7(d)/Q8(c)	2×/5	MP, M24	Missing terms [8b] · trapezoidal rule [8b] · Lagrange inverse formula [8a]
Taylor's series method: $dy/dt = e^{-2t} - 2y$, $y(0) = 0.1$, at $t = 0.1$, derivatives up to 4th degree	7	II	Q3(c)	2×/6	MP, P26	Euler on $dy/dx - 2y = 3e^x$ [3b] · coffee cooling [3d] · RK4 formula [3a]

Gauss-Seidel: $x+8y+2z=-4$; $10x+2y+z=59$; $2x-y+20z=74$	4-7	V	Q10(b)/(c)	2×/3	MP, P26	System of ODEs by matrix method [10d] · consistency conditions [10a] · Gauss elimination [10c]
Show that $x^2/(a^2+\lambda) + y^2/(b^2+\lambda) = 1$ is self-orthogonal (λ the parameter)	7	II	Q4(c)/(d)	2×/6	M23, MAE23	Modified Euler [4c] · bungee jumper Euler [4b] · analytical vs numerical [4a]
Open-top rectangular box of volume 108 cubic ft — dimensions for minimum surface area	7	I	Q1(d)/Q2(d)	2×/6	M23, MAM24	Expand about $(1, \pi/2)$ [2c] · extreme values [2b] · N-R system [1c]
Bungee jumper, $m = 68.1$ kg , $dv/dt = g - cv^2/m$, $g=9.8$, $c=0.25$ — Euler's method, steps of 1 second	4	II	Q4(b)	2×/6	M23, MAE23	Modified Euler [4c] · self-orthogonal family [4d] · analytical vs numerical [4a]
Newton-Raphson on the system $x^2+y=11$, $y^2+x=7$, from $(3.5, -1.8)$	4-7	I	Q1(c)/Q2(b)	2×/6	MP, MAE23	Lagrange multipliers [1d] · expand a^x [1b]
Find a negative root of $xe^x - \sin x = 0$ by Newton-Raphson, correct to three decimals	4-7	I	Q2(b)/(c)	2×/6	M24, P26	Maclaurin statement [2a] · expand $e^x \log x$ [2c] · distance to sphere [2d]
Write the DE of the closed circuit with L and C in series, without applied e.m.f. (M24 asks the R-C variant with source E)	2	II	Q4(a)	3×/6 identical 2×	MP, P26 (same); M24 (R-C)	Modified Euler [4c] · cardioid orthogonal trajectory [4b] · RK4 [4d]
Give the geometrical interpretation of the Newton-Raphson method	2-4	I	Q1(a)/(b)	2×/6	M23, P26	N-R system [1c] · Lagrange multipliers [1d] · series expansion [1b]
Write Lagrange's inverse interpolation formula (for two points (x_0, y_0) , (x_1, y_1))	2	IV	Q8(a)	2×/5	M23, MAM24	Missing terms [8b] · divided difference [8c] · grouped frequency [8c/d]
Obtain the expression for $\Delta^2 y_n$ (or $\nabla^2 y_n$) in terms of y values	2	IV	Q7(a)	2×/5	M23, P26	Lagrange inverse / backward difference table [7b] · grouped frequency [7c]
Write the steps involved in solving Cauchy's LDE	2	III	Q5(a)	2×/6	MP, M23	$1/(D+a)$ proof [5b] · Cauchy's equation [5c] · variation of parameters [5d]
Orthogonal trajectory of $r = b \sin \theta$ / $r = a \cos \theta$	4	II	Q3(b)/Q6(b)	2×/6	MAM24, P26	Euler's formula [3a] · cooling problem [3c] · RK4 [3d]
Euler's method on $dy/dx - 2y = 3e^x$ (MP: $y(0)=0$, $x=0.2$, $h=0.1$ P26: $y(1)=0$, $x=1.1$, $h=0.05$)	4	II	Q3(b)	2×/6	MP, P26	Taylor's series method [3c] · coffee cooling [3d] · RK4 formula [3a]
Geometrical interpretation of infinitely many solutions (MP: $2x+3y=5$, $x+1.5y=2.5$ M23: $2x+y=3$, $4x+2y=6$)	2-4	V	Q10(a)/(b)	2×/3	MP, M23	Rayleigh's power method [10b] · system of ODEs by matrix method [10d]

4 TIER 2 — Lock on the topic. Values will change.

TIER 2 — certain to appear, numbers will differ

These techniques appear in **every paper or nearly every paper**, but with fresh data each time. Drill the *method* until the setup is automatic — the arithmetic is then routine. **This is where the bulk of your marks live.**

TOPIC	MK	UNIT	SLOT	REPEATS	WHAT CHANGES / WHAT TO DRILL · APPEARS ALONGSIDE
Runge-Kutta method, 4th order	7	II	Q3(d)/Q4(d)	6×/6	In every single paper. The ODE and h change; $h=0.2$ in most. Drill the $k_1 \dots k_4$ table. <i>With:</i> orthogonal trajectories, cooling, Euler's formula [3a].
Variation of parameters	7	III	Q5(c)/(d)	5×/5	Every paper carrying Unit III. RHS is always a "hard" function — $\sec^3 x$, $\sec^2 x$, $\tan x$, e^{3x}/x^2 , $\cos(e^{-x})$. Drill the Wronskian setup. <i>With:</i> $1/(D+a)$ proof [5b] + Cauchy [5c/d].
Modified Euler's method, $h=0.2$, two iterations per stage	7	II	Q4(c)	5×/6	4 of those 5 at exactly slot 4c — the tightest positional lock in Unit II. <i>With:</i> circuit DE [4a], bungee Euler [4b], self-orthogonal or cooling [4d].
Newton's law of cooling (word problem)	4-7	II	Q3(b)/(c)/(d), Q4(d)	5×/6	Coffee, a heated object, a cooling substance. Same ODE every time. <i>With:</i> Taylor's method, RK4, orthogonal trajectories.
Taylor's series method for an ODE	4-7	II	Q3(c)/Q4(b)	5×/6	"Up to 3rd / 4th degree" — read which. <i>With:</i> Euler's method [3b], cooling [3d].
Newton's divided difference interpolation	7	IV	Q7(c)/(d), Q8(c)	5×/5	Every paper carrying Unit IV. Sometimes asks for the polynomial, sometimes a value. <i>With:</i> missing terms [8b], grouped frequency, Lagrange inverse [8a].
Maxima-minima of two variables (r-s-t test)	4-7	I	Q1(d)/Q2(b)/(d)	5×/6	x^3+y^3-3axy , $\sin x \sin y$, $\sin(x+y)$, $x^3+y^3-3x-12y+20$, $xy(1-x-y)$. <i>With:</i> expansion about a point [2c].
Taylor / Maclaurin single-variable expansion	4-7	I	Q1(b)/Q2(c)	5×/6	$\sin^{-1}x$, $\tan^{-1}x$, a^x , $\tan x$, $\sqrt{1+\sin 2x}$. <i>With:</i> N-R system [1c], Lagrange multipliers [1d].
Initial / boundary value problem (small)	4	III	Q5(b)/Q6(b)/(d)	6×/6	Always a constant-coefficient equation with 2 conditions. <i>With:</i> CF-from-roots [6a], Legendre [6c].

Orthogonal trajectories (Cartesian and polar)	2-7	II	Q3(a)/(b)/(d), Q4(a)/(c)	6x/6	Often <i>two</i> parts in one paper — a 2-mark "steps/define" plus a 7-mark problem. Learn the polar form too ($dr/d\theta \rightarrow -r^2 d\theta/dr$).
Cauchy's linear differential equation ($x^2 y'' + \dots$)	7	III	Q5(c)/(d)	4x/6	Substitution $x = e^z$. <i>With:</i> $1/(D+a)$ proof [5b], variation of parameters [5c/d].
Legendre's linear differential equation ($(ax+b)^2 y'' + \dots$)	4-7	III	Q6(c)/(d)	4x/6	Seen as $(3x+2)^2$, $(2x+1)^2$, $(2x+3)^2$, $(x+1)^2$. <i>With:</i> define linear/non-linear DE [6a], IVP [6d].
Gauss-Seidel method , 2 iterations from a given start	4-7	V	Q9(b)/Q10(b)/(c)	3x/3	Clean sweep of Unit V. Start vector given as (0,0,0) or (1,0,0) or (1,1,1). <i>With:</i> system of ODEs [10d], Gauss elimination [10c], consistency [10a].
System of ODEs by matrix method	7	V	Q10(d)	3x/3	🎯 Slot 10d in all three papers — always the very last question of the exam. Rabbit/wolf populations, a particle in a force field, a plain 2×2 system. Needs eigenvalues + eigenvectors. <i>With:</i> Gauss-Seidel [10b], Gauss elimination [10c].
Rayleigh's power method , initial vector $[1, 0, 0]^T$ or $[1, 1, 1]^T$	4-7	V	Q9(c)/(d), Q10(b)	3x/3	Iteration count varies (2 or 5). <i>With:</i> diagonalization [9c/d], define rank [9a].
Diagonalization / modal matrix (and hence A^n)	7	V	Q9(c)/(d)	3x/3	Always 7 marks. Sometimes the characteristic equation is handed to you. <i>With:</i> Rayleigh's power method [9d], rank/echelon definition [9a].
Grouped-frequency estimation (income band / age band / marks band)	7	IV	Q7(c)/Q8(c)/(d)	3x/5	The trick is building the <i>cumulative</i> frequency column first — not obvious under pressure. Drill it once. <i>With:</i> missing terms [8b], Lagrange inverse [8a].
Simpson's 3/8 rule	7	IV	Q7(d)/Q8(d)	3x/5	"6 subintervals" / "7 equidistant ordinates". <i>With:</i> divided difference, missing terms.
Numerical differentiation from a table (y' , y'')	7	IV	Q7(d)/Q8(c)	4x/5	Rate of cooling, $f''(1.1)$, $y'(\theta)$. Pick forward or backward by where x sits in the table.
Lagrange multipliers (constrained optimisation word problem)	7	I	Q1(d)/Q2(d)	4x/6	Hottest point on a sphere, open-top box, max/min distance to a sphere, divide 24 into three parts. Same method throughout.

Lagrange's interpolation (problem)	4-7	IV	Q7(b)/(c), Q8(c)	3x/5	Sometimes forward, sometimes <i>inverse</i> . <i>With</i> : divided difference, Simpson's rules.
Gauss elimination on an applied word problem	7	V	Q9(d)/Q10(c)	2x/3	Electrical network currents; dietician / nutrition mix. Write the system from the description, then eliminate.
Complementary function from given roots of the A.E.	2-4	III	Q6(a)/(b), Q5(a)	3x/6	Repeated, complex and real-distinct root cases. <i>With</i> : BVP [6b], Legendre [6c].
Trapezoidal rule	4	IV	Q8(b)	2x/5	Cheap insurance — the easiest of the three integration rules. P26 also asked for its <i>geometrical interpretation</i> (4 mk).
$\int_0^1 dx/(1+x^2)$ — same integral, different rule each year	4-7	IV	Q7(c)/Q8(b)/(d)	3x/5	Simpson 1/3 with 7 ordinates ($\rightarrow$ approximate π), Simpson 3/8 with 6 subintervals, trapezoidal with 4 strips. Know all three on this one integral.

5 TIER 3 — Probable. Seen once, or a live gap.

TIER 3 — worth an hour, not a day

Appeared once, or is on the syllabus and conspicuously *overdue*. Cover these only after Tiers 1 and 2 are solid.

ITEM	MK	UNIT	SLOT	REPEATS	NOTE
Backward-difference table construction and reading $\nabla^2 y_3$ off it	4	IV	Q7(b)	1x	M23 only. Cheap to prepare — it is just careful bookkeeping.
Newton-Gregory forward / backward interpolation as the main 7-marker	7	IV	Q7(c)	2x/5	Overdue. Sessions 19 of your portion, and the formula-statement version keeps appearing at [7a] while the full problem rarely does. A likely swap-in.
Prove $y^4 - k^4 y = 0$ has solution $C_1 \cos kx + C_2 \sin kx + C_3 \cosh kx + C_4 \sinh kx$	4	III	Q6(b)	1x	M23 only. Short and self-contained.
Consistency conditions on parameters a, b, c for solvability	7	V	Q10(c)	1x	M23 only, but rank/consistency <i>definitions</i> appear constantly — the full problem is a natural escalation.
Rank of a matrix with an unknown (find x so that rank = 2)	4	V	Q9(b)	1x	Model paper only — and the model paper has a strong record of landing in real exams.
Radius of curvature via interpolation	7	IV	Q7(d)	1x	M23 only. An interpolation problem wearing a calculus costume.
Third-order constant-coefficient LDE, e.g. $(D^3 - 3D^2 + 3D - 1)y = 2^x + x$	7	III	Q6(c)	2x/6	Tests the repeated-root CF plus a P.I. for a^x — an easily-missed standard case.

LR / RC circuit — <i>derive and solve</i> for current or charge	7	II	Q3(d)/Q4(d)	2x/6	The 2-mark "write the circuit DE" is common; the full 7-mark solve is overdue in MAC21 proper.
Newton-Cotes quadrature formula (the general result)	2-4	IV	Q7(a)/Q8(a)	0x	Never yet asked , but explicitly named in your syllabus (session 22) as the parent of Trapezoidal and Simpson. A plausible 2-marker.
Eigenvalue property proofs (e.g. λ of $A \Rightarrow 1/\lambda$ of A^{-1})	4	V	Q9(b)	1x	M24 only. Two lines if you know it, zero if you don't.

6 The 2-mark bank — 10 marks of pure recall

Every 2-mark part ever asked, by unit. You choose the easier of two each time, so knowing most of these makes all five 2-markers free.

UNIT	DEFINITIONS / STATEMENTS ASKED (REPEAT COUNT WHERE >1)
I	Geometrical interpretation of Newton-Raphson 2x · Define algebraic & transcendental equations 2x · State Maclaurin's series for one variable 2x · State Taylor's series for two variables 2x · State Taylor's series for one variable · Define maxima and minima for two variables · Explain N-R for a system of non-linear equations · Can N-R solve $x e^{-x} = 0$ near $x=2$? Justify
II	Write Euler's formula 2x · Define orthogonal trajectories 2x · DE of closed L-C circuit without e.m.f. 2x · Write RK4 formula · DE of closed R-C circuit with source E · Steps for orthogonal trajectories of $f(x, y, c) = 0$ · Steps for orthogonal trajectories in <i>polar</i> form · Two differences: analytical vs numerical methods
III	Define linear and non-linear DE with example 3x · Steps for solving Cauchy's LDE 2x · Define boundary value problem 2x · Write the CF given the roots of the A.E. 2x · Define non-homogeneous 2nd-order LDE with example · Steps for the method of variation of parameters
IV	Express $\Delta^2 y_n / \nabla^2 y_n$ in terms of y 2x · Lagrange's inverse interpolation formula 2x · State Simpson's 1/3 rule 2x · Formula for y' , y'' from Newton-Gregory forward 2x · Define forward and backward difference · Compute $\Delta^2 y_2$, $\Delta^3 y_1$ from data · First-order divided difference of a given function
V	Define rank of a matrix · Define row echelon form · Conditions for consistency / inconsistency of $AX = B$ · Define modal and spectral matrix · Is every square matrix diagonalizable? Justify · Geometrical interpretation of infinitely many solutions 2x

7 Score projection

Answering Q2 · Q4 · Q5 · Q8 · Q10 and knowing only what is on this sheet:

UNIT	QUESTION	2 MK	4 MK	7 MK	7 MK	EXPECTED
I	Q2	Maclaurin statement (T1)	$xy(1-x-y)$ max (T1)	Expand @ $(1, \pi/2)$ (T1)	Box $V=108$ / N-R root (T1)	16-20
II	Q4	L-C circuit DE (T1)	Bungee Euler (T1)	Modified Euler (T2)	Self-orthogonal / cooling (T1)	15-20
III	Q5	Steps: Cauchy's LDE (T1)	$1/(D+a)$ proof (T1)	Cauchy's equation (T2)	Variation of parameters (T2)	17-20
IV	Q8	Lagrange inverse formula (T1)	Missing terms (T1)	Divided difference (T2)	Grouped frequency (T2)	15-20
V	Q10	Consistency / modal matrix (T1)	Gauss-Seidel (T1)	Gauss elimination (T2)	System of ODEs, matrix method (T2)	14-20

Realistic band: 70–85 marks.

Tier 1 alone — the verbatim repeats — puts roughly **45–55 marks** on the table before you have solved anything new. Tier 2 carries you to the low 80s. The gap to 100 is the one unfamiliar 7-marker per unit that no prediction can eliminate.

Two failure modes to avoid.

1. Do not revise from the 2026 paper alone. It is a different scheme (25MAC21). It never tests Cauchy, Legendre, variation of parameters, Lagrange multipliers, maxima-minima, or *any* linear algebra — roughly 35 marks of your syllabus.

2. Do not skip Unit V because it is taught last. It is the smallest unit (~6 techniques), the most predictable in the whole paper, and "system of ODEs by matrix method" has been the final question of the exam in **3 out of 3** papers.

If you only get a few hours. In this order:

1. The 2-mark bank — section 6, one pass. 10 marks, pure recall.
2. $1/(D+a)$ proof + Cauchy + variation of parameters. 18 marks in Unit III, all three near-certain.
3. Unit V end to end — Gauss-Seidel, Rayleigh, diagonalization, system of ODEs by matrix method. ~6 techniques, 20 marks.
4. Modified Euler + RK4 + Newton's law of cooling. Most of Unit II.
5. Missing terms + divided difference + grouped frequency. Most of Unit IV.
6. Expand about $(1, \pi/2)$ + $xy(1-x-y)$ + the N-R system. Most of Unit I.

Compiled from 6 question papers plus the official MAC21 course handbook and session-wise portion sheet. Repeat counts are stated out of the papers that carry each unit (I–III: 6 · IV: 5 · V: 3). Predictions are inferences from observed frequency and slot position — they are not guarantees, and the paper setter is free to ignore every one of them. Treat this as a priority ordering for revision, not a substitute for it.